ABENA Alex Nelson Ryan

Looking for a 6-month end-of-studies internship in full-stack development and agentic AI (from January 2027)

Grenoble, France | +33 7 59 14 36 18 | alex.nelson.bryan@gmail.com

Portfolio | GitHub | LinkedIn

Profile

5th-year software engineering student specializing in information systems and intelligent software systems. Hands-on experience designing and implementing AI-powered enterprise software, including LLM integration, RAG pipelines with vector retrieval, AI tool execution and access controls. Works with Go, PostgreSQL, Nuxt, REST/OpenAPI and software architecture, with a strong interest in evaluating and optimizing reliable AI systems. Has solid fundamentals in algorithms and system design. Experience in AI research: co-author of three papers accepted at CERC 2026 on speech translation and recognition for low-resource African languages.

Education

Grenoble INP - ENSIMAG 2026 – 2027
Academic Exchange – Information Systems Engineering *Grenoble, France*

Institut Universitaire Saint Jean – Saint Jean Ingénieur 2022 – 2026
Computer and Information System Engineering *Yaoundé*

- GPA: 3.49/4.0 - Top of the class from first through fourth year (17.99/20 average).

Experience

Software Engineering Intern – AI & Full-Stack – TWMicronics May 2026 – Sept. 2026
MAYA Enterprise – ERP multi-tenant | Go/Gin, NuxtJS, Postgresql

- Integrated LLM and embedding models via OpenRouter for RAG; built a custom agent runtime to take control of LangChain's ReAct loop: a tool-calling loop that persists every tool call and replays it on failure, with a tool-call queue executed by workers.
- Applied tenant data isolation and human approval gates on sensitive AI tool actions; 90% of CRM actions can be run through the chatbot as well as the UI.
- Built full-stack features with Go/Gin, Nuxt 4, and PostgreSQL.

Junior Backend Developer (NestJS & MongoDB) – ITIAD Sept. 2025 – Feb. 2026
IMMOA – Plateforme immobilière

- Documented the API with Swagger/OpenAPI to enable parallel frontend and backend development.

Research & Publications

Co-author: "Three papers accepted at CERC 2026 (Galway, Ireland)" *Collaborative European Research Conference (CERC 2026), ATU Galway, Ireland*

- LinguoMT: A Zero-Shot Pilot Benchmark for African Language Speech Translation and Recognition (3rd author): built a reproducible benchmark comparing an end-to-end model and a cascade on four African languages, with validity checks that discarded invalid scores.

Projects

Lora – *Offline-First Agricultural Platform | Spring Cloud, Angular, React Native, Postgresql, Gemini* 2025 – 2026

- Architected a multi-repo Spring Cloud microservices ecosystem (Eureka, Config Server), deployed on AWS EC2.

AcadAi – *LLM-Powered Bilingual LMS Assistant (RAG + Quizzy) – Capstone Project* 2025 – 2026

- Designed a Python FastAPI RAG service with ChromaDB for bilingual institutional grounding.
- Deployed on AWS with Docker; led AI chatbot and component integration; supervised a TAM evaluation with 23 students (**91% adoption intent, mean scores $>$ 4.2**).

Skills

AI & LLM Systems	LLM integration, LLM APIs (OpenRouter, Gemini API), RAG, vector search, embeddings, pgvector, tool calling, MCP (Model Context Protocol)
Languages	Java, Python, JavaScript / TypeScript, SQL, Go, C / C++
Backend & Data	Spring Boot, Spring Cloud (microservices), Node.js (NestJS), FastAPI, Go (Gin), PostgreSQL, MongoDB, API REST, Swagger/OpenAPI, JWT
Frontend	React, Angular, Nuxt / Vue.js, React Native, HTML / CSS
Software Engineering & DevOps	UML modelling, functional specification writing, architecture decision records (ADR), unit and end-to-end testing (Playwright), acceptance testing, GitLab/GitHub CI/CD, Docker, Kubernetes, AWS (EC2, S3), Linux, Scrum, JIRA, code review, AI-assisted development (Claude, Codex) with critical review of generated code

Languages

French : Native | English : Advanced (C1 – IELTS Academic 7.5)